Tutorial-8

→ Email spam detection using decision tree

An Email Service provider wants to classify Emails spam (S) or not spam (N) using a decision tree.

Input features :-

→ Suspicious words Count

→ Contains Money Keywords (yes=1, NO=0)

→ Sender Known (yes=1, NO=0)

→ Has attachments (yes=1, NO=0)

Target variable :

Class (spam=S, Not spam=N)

Training dataset :-

Email ID	Suspicious words	Money Keywords	Sender Known	Attachment	class
1	8	1	0	1	S
2	1	0	1	0	N
3	6	1	0	1	S
4	2	0	1	0	N
5	7	1	0	1	S
6	1	0	1	1	N
7	8	1	0	1	S

8	2	0	1	0	N
9	5	1	1	0	S
10	1	0	1	1	N

Ans

1. which attritube should be selected as the root nodes. Justify your answer with calculations
2. Build the complete decision tree using the ID3 algorithm. Draw the tree structure with all nodes, branches, & leaf nodes clearly labeled
3. what is the depth of your decision tree? How many leaf nodes it have.

ID3 → only Entropy & Information Gain

Root node – IG↑

Entropy↓

$$\text{Entropy (parent)} = -\sum_{i=1}^{2} p_i \log_2 (p_i)$$

$P_{spam} = 5/10 = 0.5$ $P_{Nspam} = 5/10 = 0.5$

$\text{Spam} = 0.5 \times \log_2 (0.5) = -0.5$

$\text{Non-spam} = 0.5 \times \log_2 (0.5) = -0.5$

Entropy $= (-0.5 - 0.5) = 1$

$$S_0 = -\sum_i P_i * \log_2 (P_i)$$

$$= 0 + 1 \log_2 (1)$$

$$= 0$$

Entropy

$0 \leftarrow$ spam $= 0$

$\leftarrow$ Ns $= 5$

$S_0 = 0$

$1 \leftarrow S = 5$

$\leftarrow N = 0$

$S_1 = 0$

$$S_1 = 1 \log_2 (1) + 0$$

$$= 0$$

IGC (Money Keywords)

$$\text{Entropy (parent)} - \sum_k \frac{N_K}{N} * \text{Entropy}(K)$$

$\downarrow$ feature

$$= 1 - \left[\frac{1}{2} \times 0 + \frac{1}{2} \times 0\right] = 1$$

Entropy

$0 \rightarrow S = 2$

$\rightarrow N = 3$

$1 \rightarrow S = 3$

$\rightarrow N = 2$

$$S_0 = -\left[\frac{2}{5} \log_2 \left(\frac{2}{5}\right) + \frac{3}{5} \log_2 \left(\frac{3}{5}\right)\right]$$

$$= -\left[\frac{2}{5} \times -1.32 + \frac{3}{5} \times -0.73\right]$$

$$= 0.96$$

$S_1 = 0.96$

$= 1 - \left[\frac{1}{2} \times 0.96 + \frac{1}{2} \times 0.96\right] = 0.04$

Entropy (sender)

$0 \rightarrow S = 4$, $0 \rightarrow N = 0$

$S_0 = 0$

$1 \rightarrow S = 1$, $1 \rightarrow N = 5$

$S_1 > 0$

$IG < 0.1$

Entropy (Suspicious words)

Randomly distributed

$E > 0$

$IG \approx 1$

Root node - money Keywords

Money Keywords

0 → Ns

1 → S

Depth = 1